



COMPUTATIONAL INTELLIGENCE

MCTA 3371

MINI PROJECT:

**INTELLIGENT MOBILE ROBOT NAVIGATION USING COMPUTATIONAL
INTELLIGENCE TECHNIQUES**

SECTION 2

SEMESTER 1, 2025/2026

INSTRUCTOR:

ASSOC. PROF. DR. AZHAR BIN MOHD IBRAHIM

PROF. DR. AMIR AKRAMIN BIN SHAFIE

NAME	MATRIC NO
AMALIN BALQIS BINTI RUYUSNI	2314684
FARHA QAISARA BINTI MOHD ZAHIDI	2315994
NUR HUSNA ELYSA MAISARAH BINTI ROSLI	2310366
AUFA SIDQY BINTI MOHD SIDQY	2310542

DATE OF SUBMISSION: 24 JANUARY 2026

TABLE OF CONTENT.....	
1.0 INTRODUCTION.....	3
2.0 OBJECTIVES.....	5
3.0 METHODOLOGY.....	7
4.0 IMPLEMENTATION DETAILS.....	12
5.0 SOFT-COMPUTING MODELING.....	17
6.0 SIMULATION.....	19
7.0 FINDINGS.....	22
7.1 EVALUATION.....	22
7.2 VALIDATION AND COMPARISON.....	27
8.0 ANALYSIS OF RESULTS.....	32
9.0 CONCLUSION.....	35
AI USAGE APPENDIX.....	37
CONTRIBUTIONS.....	38

1.0 INTRODUCTION

Mobile robot navigation is a core research area in robotics and autonomous systems, focusing on enabling a robot to move from a predefined start location to a desired goal while safely avoiding obstacles in its environment. This capability is essential in many real-world applications, including indoor service robots, warehouse automation, autonomous delivery systems, and intelligent transportation. However, designing an effective navigation system is challenging due to environmental uncertainty, sensor noise, and the need for real-time decision-making without complete knowledge of the surroundings.

Conventional navigation approaches often rely on precise mathematical models, accurate sensor measurements, and deterministic control algorithms. While such methods can perform well in structured and predictable environments, they tend to struggle in complex or cluttered scenarios where uncertainty and nonlinearity are present. As a result, computational intelligence and soft computing techniques have become increasingly popular for mobile robot navigation, as they are capable of handling imprecise information and making flexible, adaptive decisions.

Computational intelligence techniques such as Fuzzy Logic, Genetic Algorithms, and Artificial Neural Networks provide powerful tools for intelligent navigation. Fuzzy Logic allows the modeling of human-like reasoning using linguistic rules, making it suitable for decision-making in uncertain environments. Genetic Algorithms offer robust optimization capabilities inspired by natural evolution, enabling the fine-tuning of controller parameters for improved performance. Artificial Neural Networks, on the other hand, provide learning and generalization abilities, allowing the system to adapt its behavior based on experience and data.

Despite their individual strengths, each of these techniques has limitations when applied independently. Fuzzy systems may suffer from suboptimal rule tuning, Genetic Algorithms alone

lack real-time decision-making, and Neural Networks may require large amounts of training data and lack interpretability. To address these challenges, hybrid computational intelligence approaches that combine multiple techniques have been proposed to enhance navigation performance and robustness.

In this project, a hybrid navigation controller integrating Fuzzy Logic, Genetic Algorithms, and Artificial Neural Networks is developed for intelligent mobile robot navigation in a simulated 2D environment. The robot is required to navigate from a start point to a goal point while avoiding obstacles placed within the environment. The hybrid controller determines the robot's movement based on sensory inputs such as distance and direction to the goal as well as proximity to obstacles. The performance of the proposed system is evaluated across different simulated maps to demonstrate its effectiveness in achieving safe, efficient, and reliable navigation.

2.0 OBJECTIVES

This project aims to investigate the effectiveness of hybrid computational intelligence techniques in solving the mobile robot navigation problem. Mobile robot navigation requires intelligent decision-making to enable a robot to move from a start location to a target destination while avoiding obstacles in uncertain and complex environments. By integrating Fuzzy Logic, Genetic Algorithms, and Artificial Neural Networks, this study seeks to develop an adaptive and robust navigation controller capable of handling imprecision, nonlinearity, and dynamic decision requirements.

The hybrid approach leverages the strengths of each technique: Fuzzy Logic provides rule-based reasoning similar to human decision-making, Genetic Algorithms optimize controller parameters for improved performance, and Neural Networks contribute learning and generalization capabilities. Through simulation-based experimentation, this project evaluates how such a hybrid system can enhance navigation efficiency, safety, and reliability across different environments. The main objectives of this project are as follows:

1. Intelligent Navigation Decision-Making

The first objective is to design an intelligent decision-making system that enables the mobile robot to determine appropriate movement actions based on sensory inputs such as distance to the goal and proximity to obstacles. The hybrid computational intelligence controller is intended to produce smooth, safe, and goal-oriented navigation behavior without relying on precise mathematical models of the robot or environment.

2. Obstacle Avoidance and Path Optimization

Another objective of this project is to ensure effective obstacle avoidance while maintaining an efficient navigation path. By using Fuzzy Logic rules optimized through Genetic Algorithms and supported by Neural Network learning, the robot is expected to minimize collisions, reduce unnecessary movements, and follow a shorter and smoother path toward the target location.

3. Performance Evaluation Across Different Environments

This project also aims to evaluate the navigation performance of the proposed hybrid controller in different 2D simulated environments with varying levels of complexity. Performance metrics such as goal-reaching success rate, path length, number of collisions, and time steps required to reach the goal are analyzed to assess the effectiveness and robustness of the controller.

4. Analysis of Hybrid Computational Intelligence Effectiveness

The final objective is to analyze the advantages of using a hybrid computational intelligence approach compared to single-method solutions. By examining navigation outcomes and performance metrics, this study aims to demonstrate how the integration of Fuzzy Logic, Genetic Algorithms, and Neural Networks leads to improved adaptability, robustness, and overall navigation performance in mobile robot systems.

3.0 METHODOLOGY

Fuzzy Logic Controller

Fuzzy logic serves as the primary navigation controller in the proposed system. It enables the robot to make movement decisions using linguistic rules rather than precise mathematical models. The fuzzy controller processes sensory information such as obstacle proximity and goal direction through predefined membership functions and if-then rules. Based on fuzzy inference, the controller determines the robot's movement at each step, allowing effective obstacle avoidance and goal-oriented behavior under uncertainty.

Obstacle Avoidance and Direction Selection Mechanism

For each candidate movement direction i , the robot measures the obstacle distance along that direction. A distance-based penalty function is applied to discourage unsafe movements. The penalty function is defined as:

$$P_{\text{obs}}(i) = \begin{cases} -\infty, & d_i < 0.8 \\ -10(1.5 - d_i), & 0.8 \leq d_i < 1.5 \\ -\frac{2(3.0 - d_i)}{1.5}, & 1.5 \leq d_i < 3.0 \\ 0, & d_i \geq 3.0 \end{cases}$$

The suitability of each movement direction is evaluated by aggregating the obstacle penalty with other heuristic components such as goal-seeking behavior and directional continuity. The total score for direction i is expressed as:

$$S_i = P_{\text{obs}}(i) + P_{\text{goal}}(i) + P_{\text{prev}}(i)$$

Finally, the robot selects the movement direction with the maximum aggregated score. This selection process serves as the defuzzification step of the fuzzy-inspired controller:

$$i^* = \arg \max_i (S_i)$$

Performance Metrics and Formulas

To objectively evaluate the effectiveness of the navigation controllers, a set of performance metrics was defined. These metrics quantify the robot's ability to reach the goal safely, efficiently, and smoothly, and they form the basis for comparing the standard Fuzzy controller with the Hybrid GA–Fuzzy–ANN controller. The metrics considered in this study include success rate, steps to goal, number of collisions, path efficiency, path smoothness, and overall performance score. Each metric is calculated using the following formulas.

1. Success Rate (%)

The success rate measures the percentage of trials in which the robot successfully reached the goal without collisions.

$$\text{Success Rate (\%)} = (\text{Number of successful trials} / \text{Total trials}) \times 100\%$$

2. Steps to Goal

This metric records the total number of discrete movement steps the robot requires to reach the goal in each trial. Lower step counts indicate more efficient navigation.

$$\text{Steps to Goal} = \text{Total steps taken during navigation}$$

3. Collision Count

The number of collisions counts how many times the robot contacts an obstacle during navigation. Zero collisions indicate safe operation.

$$\text{Collision Count} = \text{Number of invalid movement attempts}$$

4. Path Efficiency (%)

Path efficiency evaluates how directly the robot moves toward the goal, by comparing the actual distance traveled to the optimal distance.

$$\text{Path Efficiency (\%)} = (\text{Optimal distance} / \text{Actual distance traveled}) \times 100\%$$

5. Path Smoothness (%)

Path smoothness quantifies how smooth the robot's trajectory is, penalizing sharp turns or abrupt changes in direction. A higher score represents smoother navigation.

$$\text{Path Smoothness (\%)} = (1 - \text{average turn angle} / \pi) \times 100\%$$

6. Overall Performance Score (%)

The overall performance score combines the above metrics into a single measure to provide a holistic evaluation of navigation effectiveness. Weighting factors can be applied depending on the importance of each metric.

$$P = \min (100, \max(0, 40S + \min (20, E/5) + (20 - 2\min (C, 10)) + (S_m/10) + \max (0, 10 - S_t/50)))$$

Where:

- S = Success indicator (1 if goal reached, 0 otherwise)
- E = Path efficiency (%)
- C = Number of collisions
- S_m = Path smoothness (%)
- S_t = Total number of steps taken

Genetic Algorithm Optimization

A Genetic Algorithm is employed to optimize navigation-related parameters that influence the fuzzy decision-making process. GA operates as an offline optimization technique by evaluating multiple candidate solutions using a fitness function based on navigation performance metrics such as path length and efficiency. Through evolutionary operations including selection, crossover, and mutation, GA improves the quality of navigation decisions by refining parameter values that contribute to smoother and more efficient paths.

Artificial Neural Network Assistance

An Artificial Neural Network is incorporated to introduce learning capability into the navigation system. The ANN learns navigation patterns from simulation data and assists the fuzzy controller by refining movement decisions, particularly in complex environments. The ANN captures nonlinear relationships between environmental inputs and navigation behavior, thereby enhancing adaptability and improving decision consistency without replacing the fuzzy logic controller.

Why Choose a Hybrid GA–Fuzzy–ANN Approach

A hybrid GA–Fuzzy–ANN approach was chosen to address the limitations of using a single soft computing technique for mobile robot navigation. Fuzzy logic provides interpretable, human-like reasoning and handles uncertainty effectively, but its performance depends heavily on manually defined rules and membership functions. Genetic Algorithms are incorporated to systematically optimize navigation-related parameters, improving path efficiency and reducing unnecessary movements. Artificial Neural Networks introduce learning capability by capturing nonlinear relationships between environmental inputs and navigation behavior, enhancing adaptability in complex environments. By integrating fuzzy reasoning, evolutionary optimization, and learning-based refinement, the hybrid approach delivers more efficient, smoother, and robust navigation performance, particularly in environments with high complexity.

4.0 IMPLEMENTATION DETAILS

The mobile robot navigation system was implemented using Python in a simulated two-dimensional environment. The implementation focuses on integrating soft computing controllers with a grid-based navigation framework to evaluate intelligent movement and obstacle avoidance behavior. Numerical computations were handled using standard Python numerical libraries, while visualization tools were used to display robot trajectories and performance metrics throughout the simulation process.

The environment for path-following has been simulated using a 2D grid in which each grid point corresponds to either free space or an obstacle. The start and goal points are set for all experiments to maintain uniformity. Two different kinds of environments, which are a simple map consisting of few obstacles and a complex map consisting of more obstacles arranged like mazes are developed. This helps to analyze the performance of various controllers for path-following depending upon complexity.

The robot is simulated as a simple point agent which occupies one cell on the 2D grid map. For each step performed in carrying out the simulation, information pertaining to the robot's position relative to nearby obstacles and also the orientation of the goal position is collected. The collected information is then used to determine the robot's next step based on the navigation controller designed for this purpose. The robot is restricted to move and is not allowed to move into any obstacle cell to perform its actions safely and correctly.

Two types of navigation controllers, a fuzzy logic controller and a hybrid soft computing controller were designed and tested. For the fuzzy logic controller, the fuzzy logic system employed a rule-based decision support system for deciding the motion of the robot based on

approximate reasoning. Meanwhile, a hybrid soft computing approach brings together fuzzy logic, optimization, and learning components. A genetic algorithm, GA module optimizes the fuzzy logic system while an artificial neural network module helps refine the decision on navigation based on patterns discovered by learning. The hybrid method makes the system more robust for optimization and more efficient for navigation in complex environments with saturated obstacles.

The controllers were integrated into a closed-loop navigation framework, where sensing, decision-making, and movement execution occurred iteratively until the robot reached the goal or a termination condition was met. This modular implementation allowed both controllers to operate under identical conditions, ensuring a fair and consistent comparison of their navigation performance.

First of all, the fuzzy logic controller, obstacle avoidance behavior is governed by a fuzzy-inspired rule base that evaluates each candidate movement direction based on the measured obstacle distance. *Table 1* summarizes the IF–THEN rules applied during the implementation to penalize unsafe directions and encourage safe navigation.

Rule	IF	THEN
R1	Obstacle Distance is Very Near	Movement Direction is Blocked
R2	Obstacle Distance is Near	Movement Direction is Strongly Penalized
R3	Obstacle Distance is Medium	Movement Direction is Mildly Penalized
R4	Obstacle Distance is Far	Movement Direction is Not Penalized

Table 1. Each rule evaluates the suitability of a candidate movement direction based on the corresponding obstacle distance.

These rules are applied for each candidate movement direction. The robot calculates a penalty score based on the rule corresponding to the measured obstacle distance. Candidate directions with lower penalties (safer directions) are favored, and the final movement direction is selected by combining the obstacle penalty with goal-seeking and directional continuity criteria. This ensures safe and efficient navigation.

Then, for the hybrid navigation controller, a Genetic Algorithm (GA) was implemented to optimize the parameters of the fuzzy logic controller. The GA operates by evolving a population of candidate solutions, where each individual represents a set of fuzzy controller parameters. During each generation, the performance of every individual is evaluated using robot navigation simulation, and fitness will be determined based on the efficiency of navigation and the ability to avoid obstacles. Evolutionary strategies used in the optimization process include selection, crossover, mutation, and elitist strategies. The specific GA configuration parameters used in this study are summarized in *Table 2*.

Parameter	Value
Population Size	15 individuals
Number of Generations	8
Total Fitness Evaluations	120 simulations
Selection Method	Elite selection (top 4 individuals)
Crossover Method	Arithmetic crossover (average of two parents)
Mutation Method	Gaussian mutation (probability = 0.3, $\sigma = 0.1$)
Elitism Strategy	Best 4 individuals preserved

Table 2. Genetic Algorithm (GA) Configuration Parameter

The GA improves navigation performance by refining fuzzy parameters such as membership function shapes and rule weights. By systematically exploring the parameter space, the GA produces configurations that lead to smoother, more efficient paths and better obstacle avoidance.

Lastly, for the hybrid navigation controller, an Artificial Neural Network (ANN) was incorporated to refine navigation decisions through learning from simulation data. The ANN operates as a supporting module to the fuzzy logic controller by capturing navigation patterns and improving movement decisions in complex environments. The ANN configuration used in this implementation is summarized in *Table 3*.

Parameter	Description
Network Type	Feedforward Neural Network
Input Parameters	Obstacle distances and goal orientation
Output	Movement direction refinement
Role in Controller	Refinement of fuzzy logic decision
Training Data	Simulation-generated navigation data
Learning Mode	Offline supervised learning
Integration Method	Combined with fuzzy logic controller

Table 3. Artificial Neural Network (ANN) Configuration

The ANN captures nonlinear relationships between environmental inputs and optimal movement decisions. Its output is combined with the fuzzy controller's decisions to enhance navigation adaptability, allowing the robot to perform better in unfamiliar or complex scenarios.

However, the obstacle penalty computation, Genetic Algorithm (GA) fitness evaluation, and navigation performance assessment were implemented according to the mathematical formulations presented in the Methodology section. Obstacle distance measurements were used to calculate penalty values for candidate movement directions, which were combined with goal-seeking criteria to select the optimal direction. The GA optimized navigation parameters using a fitness function based on path length and efficiency, while overall navigation performance was evaluated through simulation results.

5.0 SOFT-COMPUTING MODELING

Soft computing modeling is an approach that integrates computational techniques capable of handling uncertainty, imprecision, and nonlinear complexity in real-world systems. Unlike traditional hard computing, which relies on precise inputs and rules, soft computing models simulate intelligent behavior through approximate reasoning, learning, and optimization. In this project, the proposed robot navigation system is modeled using a hybrid soft computing architecture that combines fuzzy logic, genetic algorithms, and artificial neural networks. The hybrid framework allows the robot to navigate efficiently in both simple and complex environments, providing robustness, adaptability, and smooth operation.

The fuzzy logic component serves as the core decision-making unit of the navigation system. It processes information about the environment and the goal using linguistic variables. Inputs such as obstacle proximity and goal direction are converted into linguistic values using membership functions. A predefined set of if-then rules is then applied to determine the robot's movement. Through a fuzzy inference mechanism, these rules are combined to evaluate the situation, and the resulting fuzzy outputs are converted into crisp movement commands for the robot. This allows the system to make flexible and interpretable decisions under uncertain conditions.

The genetic algorithm is used to optimize navigation parameters within the fuzzy system, such as the shapes of membership functions or rule weights. An initial population of candidate parameter sets is generated and evaluated using a fitness function that considers metrics like path efficiency and the number of steps taken. Through evolutionary operations including selection, crossover, and mutation, the population evolves over successive generations, resulting in

parameter configurations that enhance the fuzzy controller's performance. This ensures that the robot can navigate efficiently and adaptively in different scenarios.

The artificial neural network complements the fuzzy controller by learning navigation patterns from simulation or real-world data. The ANN captures nonlinear relationships between environmental inputs and navigation actions through its hidden layers. The network's output provides refined movement commands that enhance the fuzzy controller's decision-making, particularly in complex or previously unseen environments.

By integrating fuzzy logic, genetic algorithms, and neural networks, the hybrid model forms a unified navigation framework. The fuzzy logic generates the primary decision, the genetic algorithm optimizes system parameters, and the ANN refines decisions based on learned patterns. This cooperative interaction results in improved efficiency, smoothness, and robustness compared to a standalone fuzzy controller. Ultimately, the hybrid soft computing model allows the robot to navigate intelligently, avoiding obstacles and reaching its goal efficiently in various environments.

6.0 SIMULATION

This simulation focuses on the development of an intelligent mobile robot navigation system implemented in Python, based on fuzzy logic and the hybrid fuzzy approach to make decisions and avoid obstacles. Additionally, a Graphical User Interface (GUI) was created for this project to offer an interactive platform to display the navigation of the robot and the results of comparisons between various controllers.

The GUI allows users to initialize the simulation environment by selecting a predefined 2D map, which includes a start position, goal position, and static obstacles.

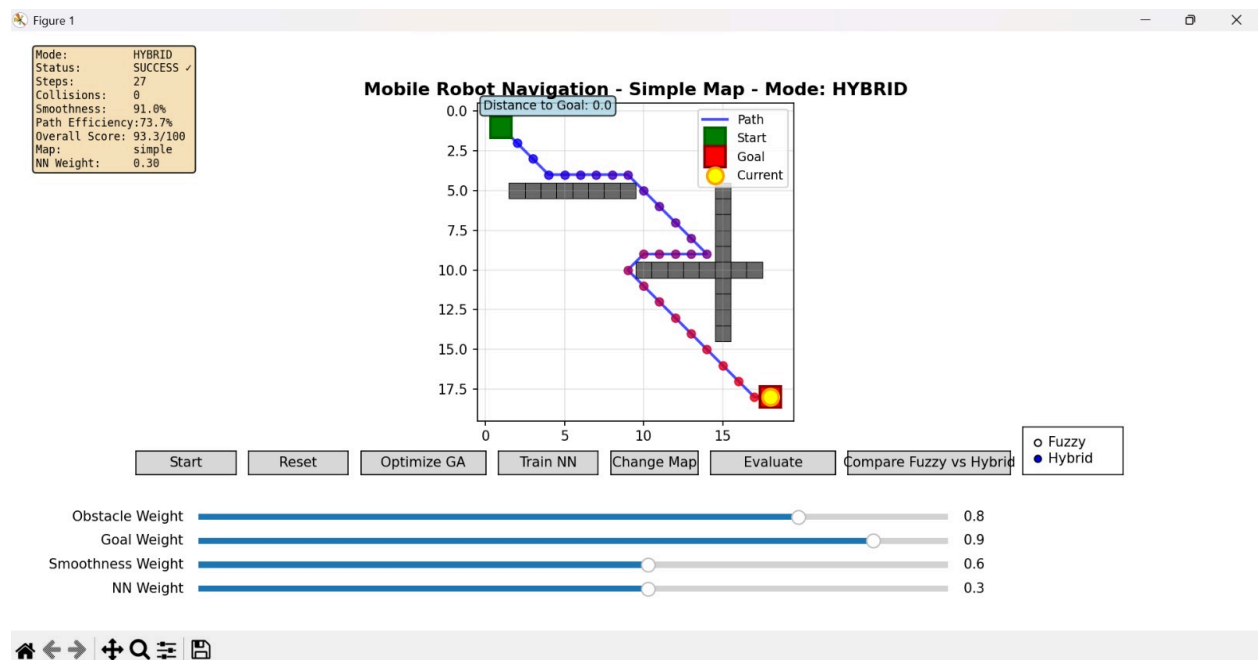


Figure 1. Graphical User Interface of Simple Grid-Based Map Displaying Robot Navigation, Obstacles, and Performance Metrics

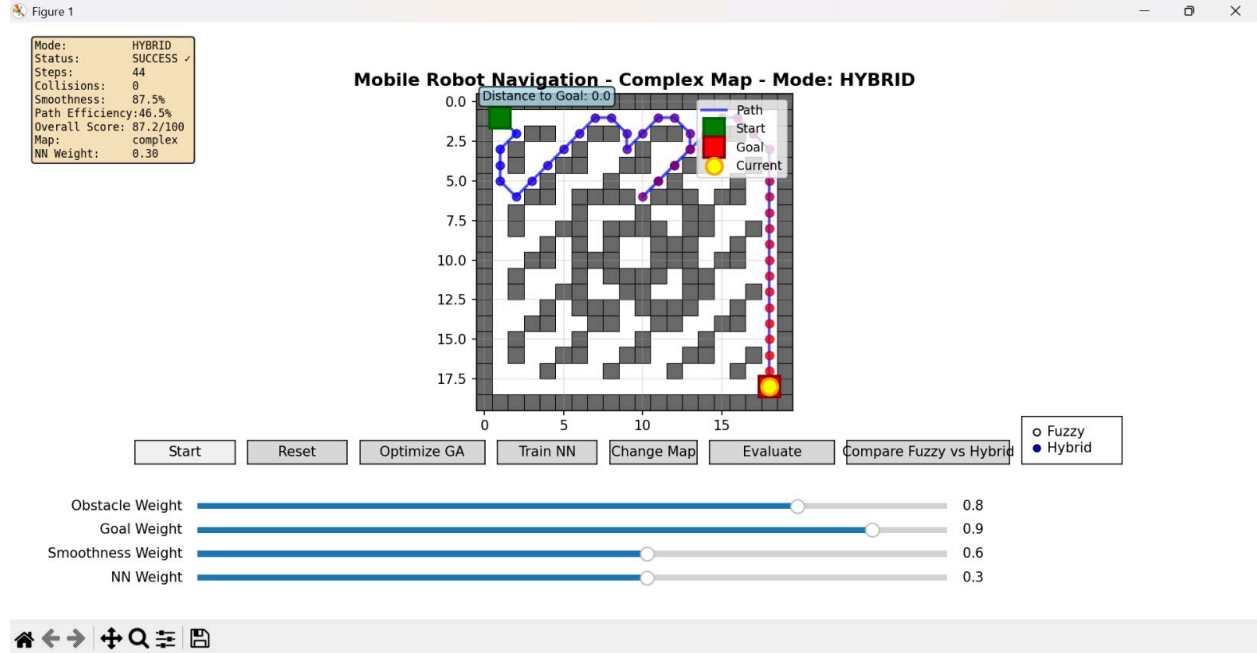


Figure 2. Graphical User Interface of Complex Grid-Based Map Displaying Robot Navigation, Obstacles, and Performance Metrics

Both maps shown in *Figure 1* and *Figure 2* were used to evaluate the navigation performance of the fuzzy-only and hybrid controllers under different levels of environmental complexity. The robot is modeled as a simple point robot that moves in discrete steps based on the outputs generated by the navigation controller. The important sensory inputs to the controller would be the distance towards the goal and the distance to the other objects, which would be processed based on fuzzy inference or hybrid logic for deciding the path towards which the robot would move.

Through this GUI, it is easy to pause, stop, or restart the simulation process by observing the real-time movement of the robot. The movements of the robot, start point, end point, obstacles, and real-time path of the robot while navigating will be displayed on this GUI. Such

visualization will help in illustrating how the controller reacts to environmental constraints to navigate the robot to its target point.

Besides the real-time display capability, another feature of the GUI is the evaluation part where it shows the performance measures of the navigation functions such as success rate of goal reaching, number of steps required, path efficiency, path smoothness, collision incidence, and total performance score. The trials can be run simultaneously and the corresponding data will be shown through comparison graphics. The comparison graphics enable the viewer to interpret the differences in performance of both fuzzy only control and hybrid control.

The simulation results show that both controllers successfully guide the robot to the goal in the simple map without collisions. The GUI also emphasizes which trial has the best performance by visualizing the optimal path taken by the robot. Along with the integration of intelligent control algorithms, the graphical interface provided by the system allows intuitive and fast analysis of the behavior of a robot navigating and validates the effectiveness of soft computing and hybrid approaches in autonomous navigation tasks.

7.0 FINDINGS

The navigation controllers were evaluated across 5 trials on a simple map environment. Performance was measured using five key metrics: success rate, steps taken, collision count, path efficiency, path smoothness, and an overall performance score. The results reveal distinct performance profiles for each controller, which are fuzzy and hybrid methods.

7.1 EVALUATION

An evaluation of both controllers was conducted separately on simple and complex map environments.

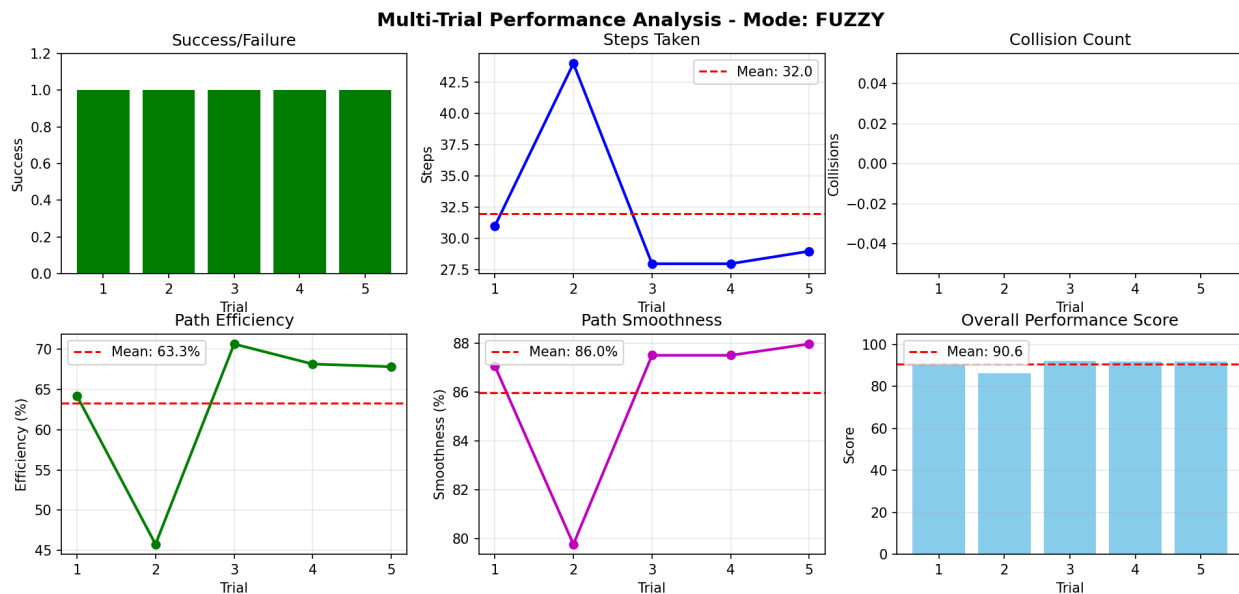


Figure 3: Evaluation of Navigation Controllers Using Fuzzy Mode on Simple Map

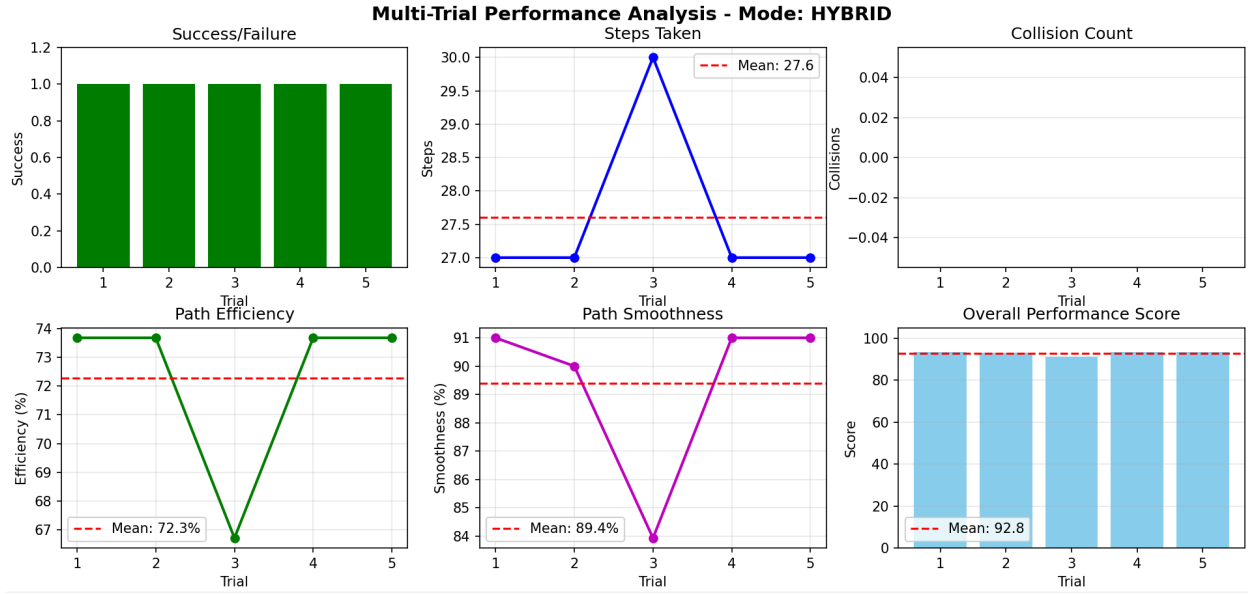


Figure 4: Evaluation of Navigation Controllers Using Hybrid Mode on Simple Map

Based on **Figure 3 and Figure 4**, both the standard Fuzzy controller and the Hybrid controller demonstrated excellent reliability in the simple map environment, achieving a 100% goal-reaching success rate with zero collisions. This indicates that both controllers are fundamentally safe and capable of basic navigation in low-complexity environments.

Despite this similarity in reliability, notable differences are observed in navigation efficiency and path quality. The standard Fuzzy controller required a higher mean number of steps (32) to reach the goal, indicating less optimal decision-making. Its path efficiency was recorded at 63.3%, with a smoothness score of 86%, resulting in an overall performance score of 90.6%. These results suggest that while the Fuzzy controller is robust and consistent, it tends to produce longer and less optimized paths.

In contrast, the Hybrid controller exhibited superior performance across all efficiency-related metrics. It reached the goal using fewer steps, with a mean of 27.6, demonstrating faster convergence. Additionally, it achieved higher path efficiency of 72.3% and improved path smoothness at 89.4%. These improvements collectively contributed to a higher overall performance score of 92.8%.

In the simple map environment, both controllers achieved a 100% success rate with zero collisions, confirming reliable basic navigation. However, the Hybrid controller demonstrated improved efficiency, requiring fewer steps and producing smoother and more efficient paths than the standard Fuzzy controller.

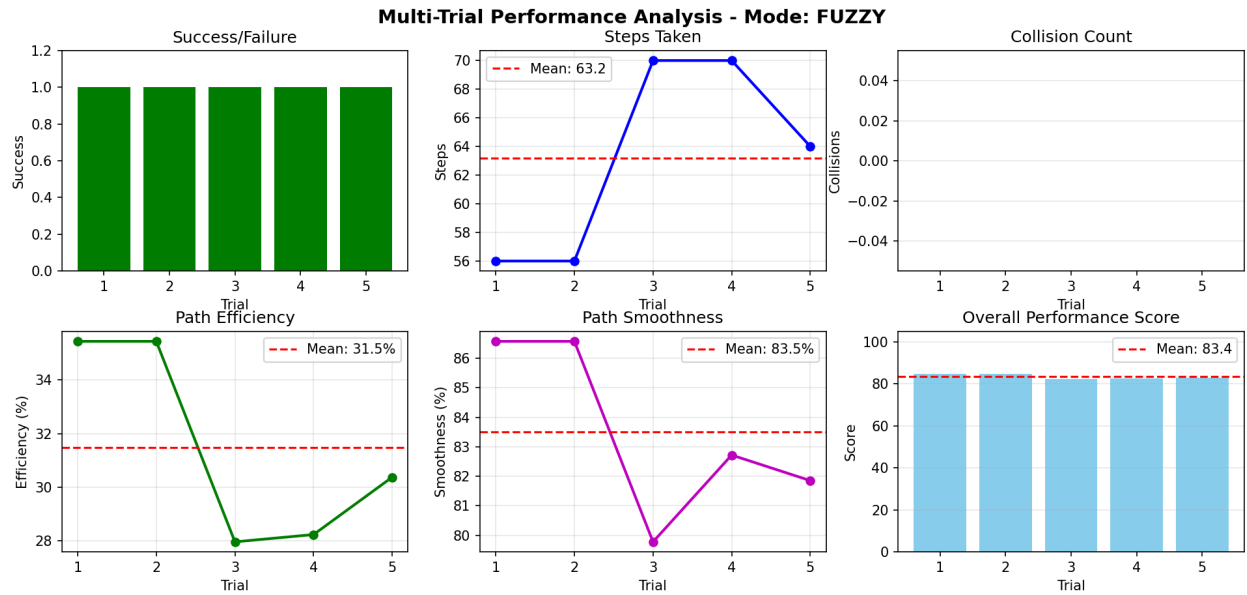


Figure 5: Evaluation of Navigation Controllers Using Fuzzy Mode on Complex Map

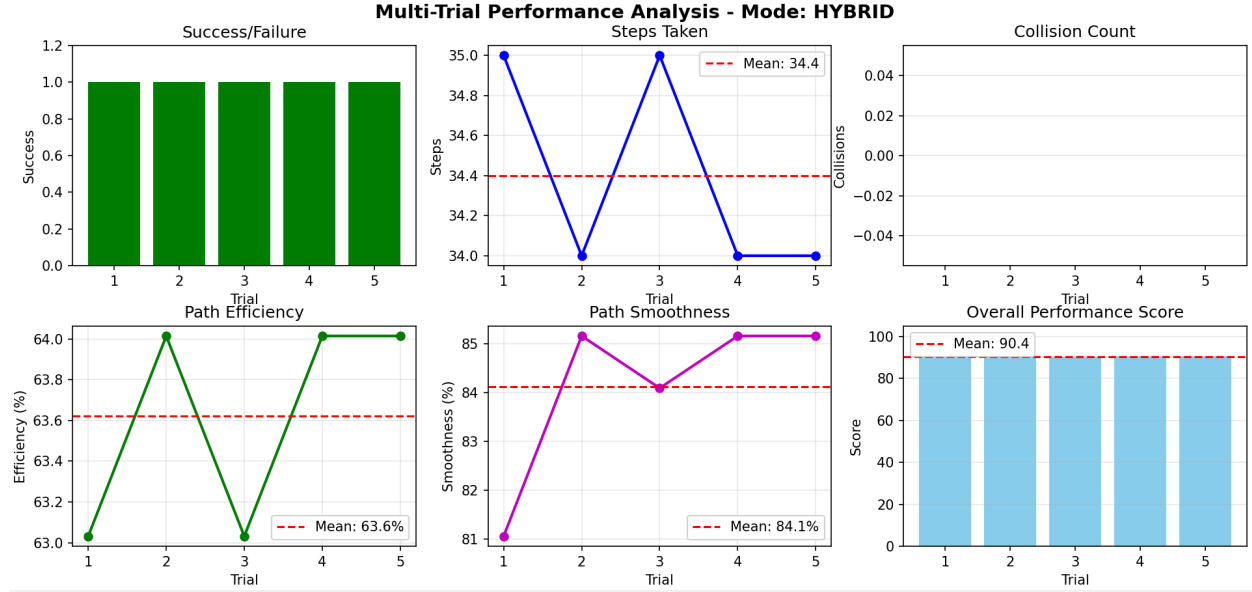


Figure 6: Evaluation of Navigation Controllers Using Hybrid Mode on Complex Map

In the more challenging complex map environment, the performance gap between the two controllers widened considerably. As shown in *Figure 5* and *Figure 6*, both controllers maintained perfect reliability, achieving a 100% success rate with zero collisions. This confirms that both approaches remain safe even under increased environmental complexity.

However, the standard Fuzzy controller experienced a significant decline in efficiency-related performance metrics. It required a mean of 63.2 steps to reach the goal, while its path efficiency dropped sharply to 31.5%. Despite maintaining a relatively high path smoothness of 83.5%, its overall performance score decreased to 83.4%. These results indicate that the Fuzzy controller struggled to generate optimized paths when navigating complex obstacle configurations.

In contrast, the Hybrid controller demonstrated strong robustness and adaptability. It reached the goal using substantially fewer steps, with a mean of 34.4, and achieved a significantly higher path efficiency of 63.6%. The controller also maintained high path smoothness at 84.1%. As a result, it attained a much higher overall performance score of 90.4%. This superior performance highlights the effectiveness of the Hybrid controller's adaptive learning mechanisms in handling complex navigation tasks, particularly in optimizing path planning under challenging conditions.

7.2 VALIDATION AND COMPARISON

The Fuzzy and Hybrid controllers were directly compared using a single graph to evaluate their relative performance and learning progression over time.

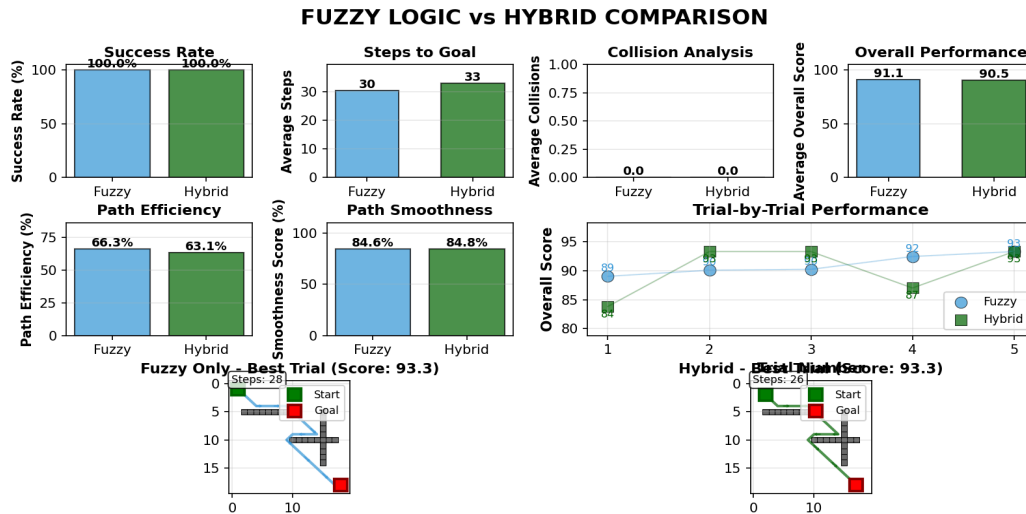


Figure 7: Comparison of Fuzzy and Hybrid Mode on A First Trial on Simple Map

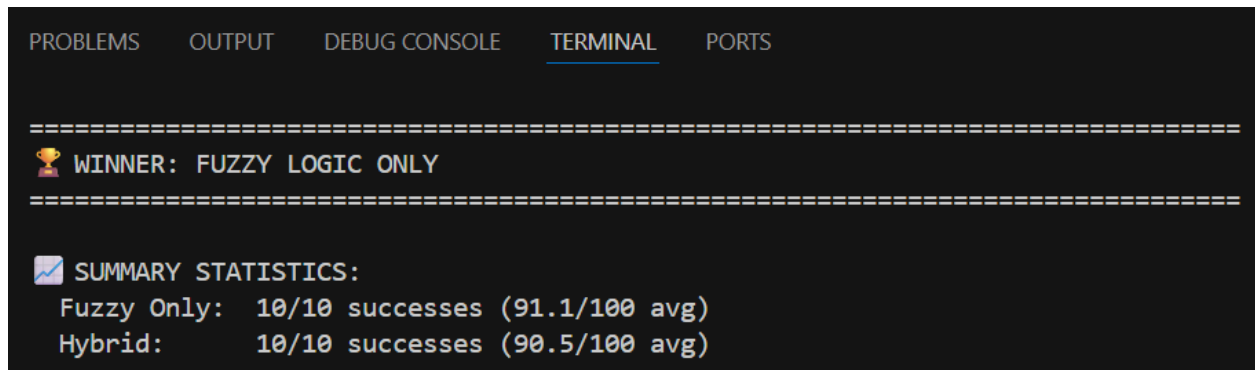


Figure 8: Summary Statistics of Fuzzy Logic and Hybrid Navigation Performance on First Trial on Simple Map

Based on **Figure 7** and **Figure 8**, during the first trial on the simple map, both controllers demonstrated excellent performance, achieving a 100% success rate with zero collisions. The Fuzzy controller exhibited slightly higher initial efficiency, reaching the goal in 30 steps. It

achieved a higher path efficiency of 66.3% and a path smoothness of 84.6%, resulting in a marginally superior overall performance score of 91.1%.

In comparison, the Hybrid controller, while equally reliable, was less efficient in its initial attempt. It required 33 steps to reach the goal, with a path efficiency of 63.1% and a smoothness score of 84.8%, leading to an overall performance score of 90.5%. Both the Fuzzy and Hybrid controllers maintained zero collisions throughout the trial, confirming safe navigation behavior.

The trial-by-trial performance graph further illustrates the learning behavior of both controllers. The Fuzzy controller's performance score increased steadily during the first trial, reflecting its consistent rule-based decision-making. In contrast, the Hybrid controller's performance exhibited noticeable fluctuations, indicating that it was still adjusting and refining its navigation strategy during the early learning phase.

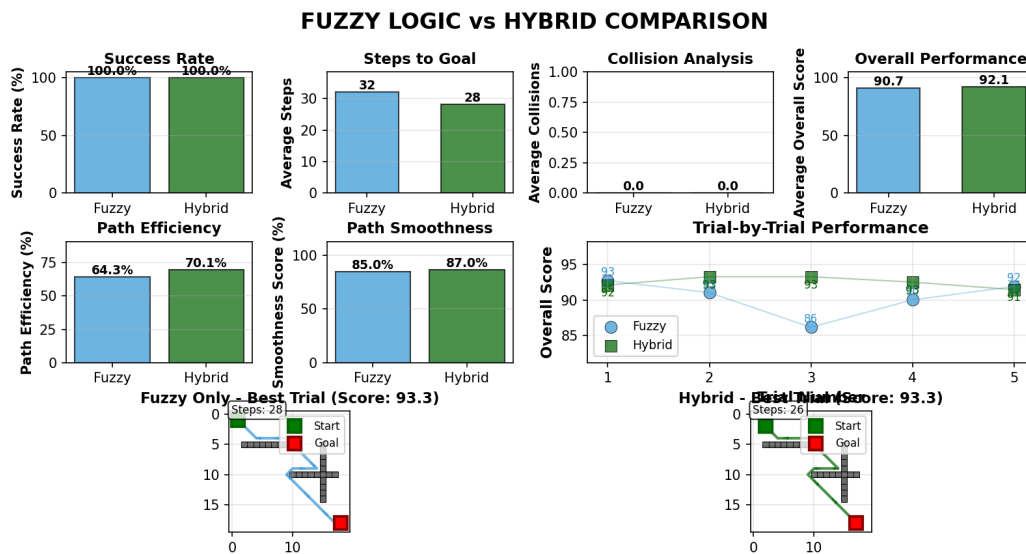


Figure 9: Comparison of Fuzzy and Hybrid Mode on A Second Trial on Simple Map

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

=====
🏆 WINNER: HYBRID (Fuzzy + GA + NN) by 1.6%
=====

📊 SUMMARY STATISTICS:
Fuzzy Only:  10/10 successes (90.7/100 avg)
Hybrid:      10/10 successes (92.1/100 avg)
```

Figure 10: Summary Statistics of Fuzzy Logic and Hybrid Navigation Performance on Second Trial on Simple Map

A clear shift in performance was observed during the second trial on the simple map. Based on *Figure 9 and Figure 10*, both controllers once again achieved perfect reliability, recording a 100% success rate with zero collisions. However, the Hybrid controller demonstrated a significant improvement compared to its first trial.

The Hybrid controller reached the goal in only 28 steps, outperforming the Fuzzy controller, which required 32 steps. Its path efficiency increased to 70.1%, surpassing the Fuzzy controller's 64.3%. In addition, the Hybrid controller achieved slightly higher path smoothness at 87%, compared to 85% for the Fuzzy controller. These improvements resulted in a higher overall performance score of 92.1% for the Hybrid controller, compared to 90.7% for the Fuzzy controller.

The performance graph for the second trial further illustrates this improvement. The Hybrid controller's performance scores became more stable with reduced fluctuations, indicating that its decision-making process had become more consistent and reliable after gaining experience. In contrast, although the Fuzzy controller maintained high performance, its score exhibited some variability, with a decrease followed by a subsequent increase. This behavior reflects the fixed-rule nature of the Fuzzy controller, which lacks adaptive learning capability.

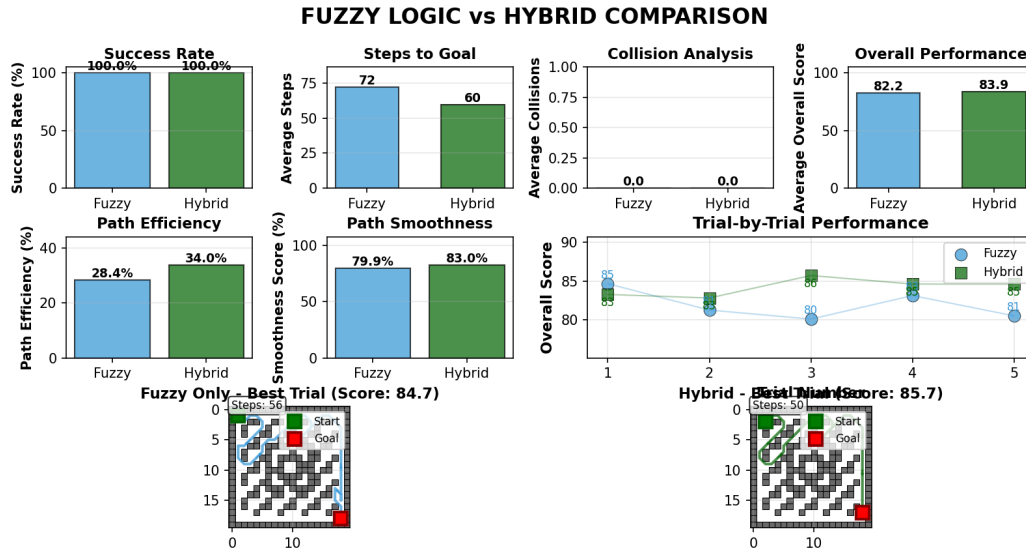


Figure 11: Comparison of Fuzzy and Hybrid Mode on Complex Map

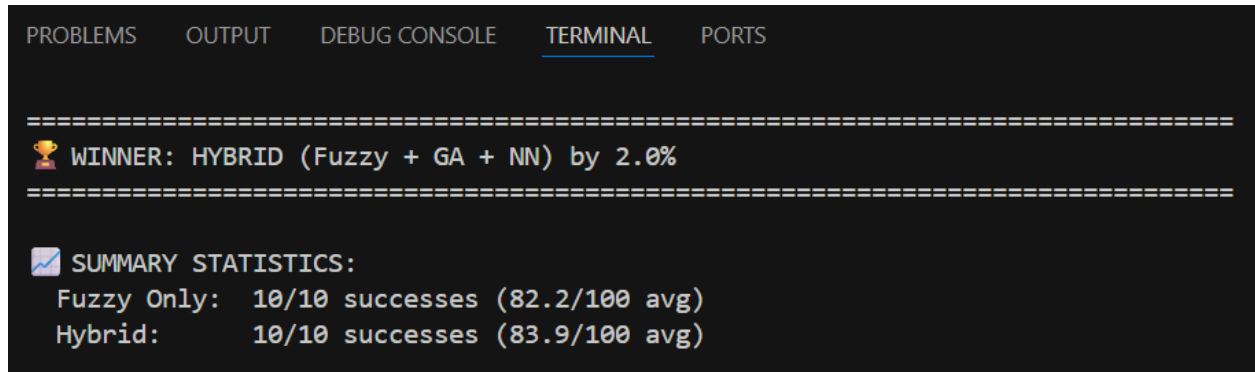


Figure 12: Summary Statistics of Fuzzy Logic and Hybrid Navigation Performance on Complex Map

Based on *Figure 11* and *Figure 12*, in the complex map environment, both controllers demonstrated robust performance by maintaining a 100% success rate with collision-free navigation. Despite this shared reliability, a clear difference in path efficiency and optimization emerged between the two approaches.

The standard Fuzzy controller, while reliable, required 72 steps to reach the goal and achieved a relatively low path efficiency of 28.4%, indicating a longer and less direct navigation path. In contrast, the Hybrid controller completed the task more efficiently, requiring only 60 steps and achieving a higher path efficiency of 34%. The Hybrid controller also demonstrated superior path smoothness at 83%, compared to 79.9% for the Fuzzy controller. As a result, the Hybrid controller attained a higher overall performance score of 83.9%, compared to 82.2% for the standard Fuzzy controller.

The trial-by-trial performance graph further highlights these differences. The Fuzzy controller's performance scores exhibited noticeable variability, with inconsistent increases and decreases, suggesting that its fixed rule-based approach struggled to consistently optimize navigation in the complex maze environment. In contrast, the Hybrid controller's performance curve was more stable and showed a slight upward trend, reflecting its adaptive learning capability and improved consistency across repeated trials.

This behavior mirrors the pattern observed in the simple map experiments, confirming that once experience is gained through initial trials, the Hybrid controller consistently outperforms the standard Fuzzy system. These results demonstrate that in challenging environments, the Hybrid controller's learning components provide a measurable advantage in producing smoother, more efficient, and more optimized navigation paths.

8.0 ANALYSIS OF RESULTS

The simulation results reveal clear differences in behavior characteristics, efficiency, and adaptability between the standard Fuzzy controller and the Hybrid GA–Fuzzy–ANN controller across both simple and complex map environments. While both controllers consistently achieved a 100% success rate with zero collisions, the Hybrid controller demonstrated superior efficiency, path quality, and learning progression.

Performance in Simple Map Environment

In the simple map, both controllers successfully navigated to the goal safely. The Fuzzy controller produced stable and smooth paths from the first trial, reflecting the consistency of its rule-based decision-making. However, its efficiency was limited by fixed rules, resulting in longer paths and moderate step counts.

The Hybrid controller initially performed slightly worse in the first trial, showing fluctuations in its performance scores. This variability reflects an early adaptation phase, where the integrated learning component, which is ANN and GA-optimized fuzzy parameters, were still refining navigation strategies. By the second trial, the Hybrid controller achieved higher efficiency and smoother paths than the Fuzzy controller, with more consistent performance. These improvements illustrate the effectiveness of GA-based parameter optimization in fine-tuning the fuzzy controller and the ANN's learning ability in adapting decisions based on previous experience.

Learning Progression and Adaptability

The trial-by-trial graphs emphasize a key distinction, which is the Fuzzy controller maintained steady but largely unchanging performance, due to its fixed rules. In contrast, the Hybrid controller displayed initial fluctuations followed by stabilization and gradual improvement, indicating successful learning and optimization. This adaptive behavior enables the Hybrid controller to refine its navigation strategy over successive trials, reduce unnecessary movements, and improve overall efficiency.

Performance in Complex Map Environment

The differences became more pronounced in the complex map environment. Although both controllers remained safe and collision-free, the Fuzzy controller required substantially more steps and exhibited lower path efficiency, with irregular score fluctuations. This highlights the limitations of fixed-rule reasoning in more challenging, maze-like scenarios.

The Hybrid controller, in comparison, consistently achieved shorter paths, higher path efficiency, and smoother trajectories. Its trial-by-trial performance was more stable and showed slight upward trends, demonstrating the combined effect of GA-optimized fuzzy parameters and ANN-guided decision refinement. These adaptive mechanisms allowed the Hybrid controller to better handle environmental complexity and consistently produce efficient navigation paths.

Overall Effectiveness of Hybrid Approach

Across both environments, the results confirm that while the Fuzzy controller is reliable and safe, it is inherently limited by its static rule base. The Hybrid GA–Fuzzy–ANN controller, by leveraging parameter optimization and learning, not only improves efficiency and path quality but also adapts to complex conditions over repeated trials. This demonstrates that integrating fuzzy reasoning, evolutionary optimization, and neural learning provides a clear advantage in intelligent mobile robot navigation.

9.0 CONCLUSION

In conclusion, this project has successfully demonstrated the application of a hybrid computational intelligence approach for intelligent mobile robot navigation in a simulated environment. By integrating Fuzzy Logic, Genetic Algorithms, and Artificial Neural Networks, the proposed navigation controller is able to guide a mobile robot from a start location to a target destination while effectively avoiding obstacles.

The use of Fuzzy Logic enables intuitive and rule-based decision-making under uncertainty, closely resembling human reasoning in navigation tasks. Genetic Algorithms contribute by optimizing controller parameters, leading to improved navigation efficiency, smoother paths, and reduced collision occurrences. Meanwhile, Artificial Neural Networks enhance the system's adaptability by providing learning and generalization capabilities, allowing the controller to perform consistently across different environments and navigation scenarios.

Simulation results indicate that the hybrid controller performs effectively in both simple and complex maps, achieving a high goal-reaching success rate while maintaining safe distances from obstacles. Compared to single-method approaches, the hybrid system demonstrates improved robustness, adaptability, and overall navigation performance. These findings highlight the advantages of combining multiple computational intelligence techniques to address the challenges of autonomous navigation without relying on precise mathematical models or complete environmental knowledge.

Although the simulation-based implementation provides promising results, further improvements can be made by extending the system to handle dynamic obstacles, real-time learning, and more complex environments. Future work may also involve implementing the

hybrid navigation controller on a physical robot platform to evaluate its performance in real-world conditions. Overall, this project emphasizes the effectiveness and practicality of hybrid computational intelligence techniques for solving complex mobile robot navigation problems and their potential applications in autonomous robotic systems.

AI USAGE APPENDIX

AI tools including ChatGPT, Claude, and DeepSeek were employed in this project for multiple supportive roles including concept clarification regarding fuzzy logic and hybrid system fundamentals, generating initial Python code structures for controllers and GUI components, debugging syntax errors in visualization code, and enhancing the technical explanations within our report.

Key prompts used included:

- *“Generate a Python script for a mobile robot navigation simulation using fuzzy logic for obstacle avoidance.”*
- *“Propose a hybrid approach combining fuzzy logic, genetic algorithm, and neural network for mobile robot navigation.”*
- *“How can fuzzy and hybrid controllers be evaluated using navigation performance metrics?”*
- *“Suggest ways to visualize robot navigation paths and performance results in a Python GUI.”*
- *“Explain why the robot gets stuck in local minima and suggest improvements.”*

All AI-generated outputs were carefully reviewed, tested in our simulation environment, and modified to meet our specific requirements. Code snippets were integrated into our larger system and all implementations were validated through extensive testing. We confirm complete understanding of the final implementation and take full responsibility for all results and conclusions presented in this report.

CONTRIBUTIONS

NAME	TASKS
AMALIN BALQIS BINTI RUYUSNI (2314684)	1. FINDINGS 2. ANALYSIS OF RESULTS 3. CODING
FARHA QAISARA BINTI MOHD ZAHIDI (2315994)	1. IMPLEMENTATION OF DETAILS 2. SIMULATION 3. CODING
NUR HUSNA ELYSA MAISARAH BINTI ROSLI (2310366)	1. INTRODUCTION 2. OBJECTIVES 3. CONCLUSION 4. CODING
AUFA SIDQY BINTI MOHD SIDQY (2310542)	1. METHODOLOGY 2. SOFT-COMPUTING MODELING 3. CODING